

PARTS LIST

Semiconductors:

IC1	- 7812, +12V voltage regulator
IC2	- 7912, -12V voltage regulator
D1-D4	- 1N4007 rectifier diode
LED1-LED2	- 5mm red LED
LED3-LED4	- 5mm green LED
ZD1	- 3.1V, 0.5W zener diodes

Resistors (all $\frac{1}{4}$ -watt, $\pm 5\%$ carbon, unless stated otherwise):

R1	- 470-ohm, 0.5W
R2-R4	- 1-kilo-ohm
R5-R6	- 2.2-kilo-ohm

Capacitors:

C1-C2	- 220 μ F, 35V electrolytic
C3-C6	- 100nF ceramic disk

Miscellaneous:

X1	- 230V AC primary to 15V-0-15V, 250mA secondary transformer
S1	- On/off switch
S2	- Double-pole double-throw (DPDT) switch
F1	- 250mA fuse
ZIF1	- 8-pin IC socket (dual-in-line type)

terminal regulator. Capacitors C3, C4, C5 and C6 are decoupling capacitors. These are used to decouple DC power supplies for high-frequency noise, if any, present on the DC power supply line. Resistor R1 and zener diode ZD1 generate 3.1V DC, which is used as a

test input to the op-amp.

The working of the circuit is simple. Put the IC in socket ZIF1. Switch on the power supply through switch S1. Now toggle switch S2 and check the status of LED1 and LED2. If LED1 glows when you toggle up and LED2 glows when you toggle down, the op-amp under test is working well in both inverting and non-inverting modes.

Construction and testing

An actual-size, single-side PCB for the op-amp tester is shown in Fig. 2 and its component layout in Fig. 3. After assembling the circuit on a PCB, enclose it in a suitable box.

For testing, follow these steps:

1. Switch on the test gadget through switch S1. Measure the regulated +12V, -12V and +3V DC at test points TP1, TP2 and TP3 with respect to TP0 as shown in the test points table. Ensure that all the DC voltages are available. LED3 and LED4 indicate

correct +12V and -12 supply.

2. Toggle up switch S1.

3. Switch off the power supply.

4. Insert the op-amp (take a good op-amp) in the 8-pin zif socket (ZIF1) meant for the purpose.

5. Switch on the power supply. LED1 will glow to indicate that the op-amp is functioning properly in inverting amplifier configuration. LED2 should not glow while switch S1 is in this position.

6. Toggle down DPDT switch. LED1 will now stop glowing and LED2 will start glowing. This confirms operation of the op-amp in non-inverting mode.

7. Switch off the power supply.

8. Remove the op-amp from ZIF1 test socket.

Remember that even if the op-amp works in either of these configurations, it should still be rated as a bad IC. ●

The author is a regular contributor to EBY and has many articles published to his credit in India and abroad



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PRODUCT RANGE

Metalised Polyester & Polypropylene Capacitors

0.001 to 22 mfd., 63 to 3000 VDC,
250 VAC to 440 VAC,
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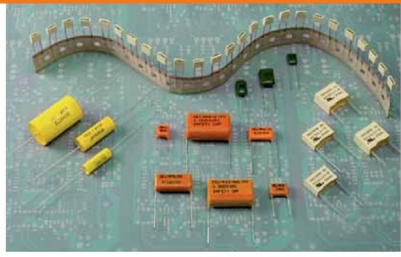
Plain (Inductive) Polyester & Polypropylene Capacitors

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APPLICATIONS

- EMV/RFI (X2 & Y2)
- Lighting (CFL, Ballast, Ignitor)
- Energy Meter
- Fan Regulator (Safety & Normal)
-and many more

APPROVALS & CERTIFICATION



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